

Coordinates conversion in acoustic positioning

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1 General information

During an acoustic positioning by ultra-short baseline transceiver a direction to a target is measured as a round-trip time. When sound speed is known a distance between USBL transceiver and transponder can be calculated. In combination with measured direction it allows to calculate target coordinates in USBL-frame (or USBL-Frame) $\vec{X}_{ut}$ in **USBLLONG**.

In some scenarios USBL transceiver can't measure distance to target. In that case bearing θ_{ut} and elevation φ_{ut} angles are provided in **USBLANGLES**. Covertion to USBL-Frame when distance d is known is:

$$\vec{X}_{ut} = d \cdot \begin{bmatrix} \sin \theta_{ut} \cos \varphi_{ut} \\ \cos \theta_{ut} \cos \varphi_{ut} \\ \sin \varphi_{ut} \end{bmatrix} \quad (1.1)$$

USBL-Frame corrdinates $\vec{X}_{ut}$ should be converted to Vessel-Frame coordinates $\vec{X}_{vt}$ by correction to USBL location $\vec{X}_{vu}$ and rotation $[R_{vu}]$:

$$\vec{X}_{vt} = \vec{X}_{vu} + [R_{vu}] \cdot \vec{X}_{ut}. \quad (1.2)$$

Vessel-Frame coordinates of target $\vec{X}_{vt}$ could be motion-compensated by using information about attitude and heading of a vessel $[R_{vn}]$. Let φ_{vn} (roll), θ_{vn} (pitch) and ψ_{vn} (yaw) angles are provided by vessels AHRS. As result Topocentric coordinates (North-East-Down) with Vessel-Frame origin are computed:

$$\vec{N}_{vt} = [R_{vn}] \cdot \vec{X}_{vt} \quad (1.3)$$

When Geodetic coordinates of Vessel-Frame origin $\vec{G}_v$ and Topocentric coordinates of target $\vec{N}_{vt}$ are known, Geodetic coordinates of target $\vec{G}_t$ can be computed.

2 Definitions

2.1 Coordinate systems

Vessel-Frame is cartesian coordinate system with origin somewere on a vessel. Usually on stern. X_v -axis is directed forward, Y_v -axis directed astarboard and Z_v -axis is directed down by right-hand rule.

USBL-Frame coordinate system has origin in the middle of USBL transceiver. Y_u -axis is directed to white mark on USBL-head (forward), Z_u -axis is directed to subconn port direction (up) and X_u by right-hand rule directs right.

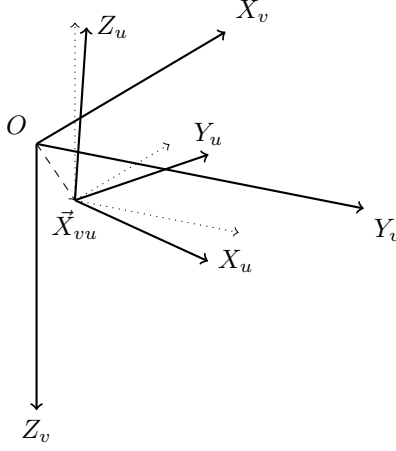


Figure 2.1: USBL-Frame located in Vessel-Frame

2.2 Rotation

Rotation of coordinates around specific axis on specific angle could be done by using Quaternions or Direction-Cosine Matrices (or Rotation Matrices).

Quaternions allows to apply rotation around arbitrary axes. Positive direction of rotation is a clockwise rotation when watching towards the axis. DCM allows the same, but its more intuitive to describe vector operations with it.

Multiplying rotation Matrix to a column vector produces rotated column vector.

DCM can be built from quaternion in case of arbitrary axis $\vec{r}$ and angle Ω or as a combination of rotations around X , Y and Z axes (in $X - Y - Z$ order).

Angles φ , θ and ψ are also known as Euler angles. Rotation matrix $[R(\varphi, \theta, \psi)] = [R_z(\psi)] \cdot [R_y(\theta)] \cdot [R_x(\varphi)]$.

Applying rotation $[R_{ab}]$ and shift $\vec{X}_{ab}$ to point $\vec{X}_a$:

$$\begin{bmatrix} x_b \\ y_b \\ z_b \end{bmatrix} = \begin{bmatrix} x_{ab} \\ y_{ab} \\ z_{ab} \end{bmatrix} + \begin{bmatrix} \cos \psi_{ab} & -\sin \psi_{ab} & 0 \\ \sin \psi_{ab} & \cos \psi_{ab} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos \theta_{ab} & 0 & \sin \theta_{ab} \\ 0 & 1 & 0 \\ -\sin \theta_{ab} & 0 & \cos \theta_{ab} \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi_{ab} & -\sin \varphi_{ab} \\ 0 & \sin \varphi_{ab} & \cos \varphi_{ab} \end{bmatrix} \cdot \begin{bmatrix} x_a \\ y_a \\ z_a \end{bmatrix}. \quad (2.1)$$

Single formula to computation rotation matrix (2.1) from Euler angles:

$$[R(\varphi, \theta, \psi)] = \begin{bmatrix} \cos \theta \cos \psi & \sin \varphi \sin \theta \cos \psi - \cos \varphi \sin \psi & \cos \varphi \sin \theta \cos \psi + \sin \varphi \sin \psi \\ \cos \theta \sin \psi & \sin \varphi \sin \theta \sin \psi + \cos \varphi \cos \psi & \cos \varphi \sin \theta \sin \psi - \sin \varphi \cos \psi \\ -\sin \theta & \sin \varphi \cos \theta & \cos \varphi \cos \theta \end{bmatrix}. \quad (2.2)$$

Angles can be extracted from DCM by:

$$\begin{aligned} \varphi &= \arctg R_{32}/R_{33}, \\ \theta &= -\arcsin R_{31}, \\ \psi &= \arctg R_{21}/R_{11}. \end{aligned} \quad (2.3)$$

DCMs has a property: $[R]^{-1} = [R]^T$.

From physical point of view, φ , θ and ψ angles are also called as Roll, Pitch and Yaw. Usually AHRS sensors provides angles for NED-frame: Roll — port up, Pitch — bow up, Yaw — heading for Vessel-Frame specified earlier.

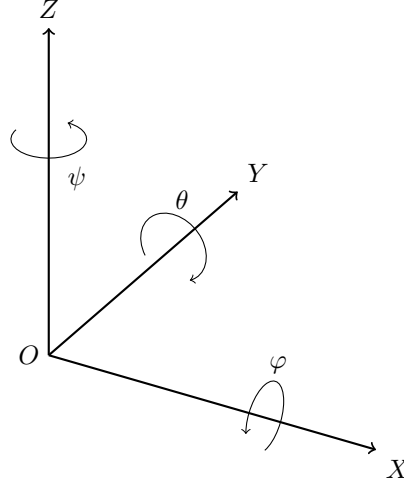


Figure 2.2: Euler angles

3 Coordinates conversion

3.1 Topocentric coordinates

As mentioned in (1.2) an offset and rotation of USBL relative to Vessel-frame is required.

It is possible that AHRS sensor located with some bias $[R_{va}]$. In this case Euler angles (or rotation matrix $[R_{an}]$) conversion is needed.

As mentioned in (1.3), coordinates in AHRS-Frame can be converted to NED by multiplication to $[R_{vn}]$. To convert coordinates from Vessel-Frame to AHRS Frame a multiplication to $[R_{va}]^{-1}$ is needed.

Inverted DCM $[R_{va}]^{-1}$ has an easy explanation. Since $[R_{vu}]$ (USBL-Frame rotation) and $[R_{va}]$ (AHRS-Frame rotation) are both specified in Vessel-Frame, first is used to convert $USBL \rightarrow VF$, second for $AHRS \rightarrow VF$. Since transformation $VF \rightarrow AHRS$ is needed, rotation matrix is inverted.

$$[R_{vn}] = [R_{an}] \cdot [R_{va}]^{-1} \quad (3.1)$$

3.2 Geographic coordinates

When GNSS is used, geodetic coordinates $\vec{G}_g$ of antenna (which is located at $\vec{X}_{vg}$ in Vessel-Frame) are measured:

$$\vec{G} = [\varphi, \lambda, h]^T \quad (3.2)$$

Geodetic Latitude φ and Longitude λ will be interpreted as in radians and in WGS-84 datum.

All geodetic computations are done in Earth-centered Earth-fixed (ECEF) coordinates. Origin of ECEF is located in center of mass of Earth. X-axis is directed to Greenwich meridian ($\varphi = 0, \lambda = 0$), Y-axis is directed to North-pole ($\varphi = \pi$) and meters are used as units.

Conversion between Geodetic and ECEF coordinates are done by EPSG:9602. Let ellipsoid curvature on specified Latitude is:

$$\nu = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}} \quad (3.3)$$

where a – major semiaxis and e – eccentricity of WGS-84 ellipsoid. These constants can be found in table 3.1.

Geodetic coordinates conversion to ECEF is:

$$\begin{aligned} X &= (\nu + h) \cdot \cos\varphi \cdot \cos\lambda \\ Y &= (\nu + h) \cdot \cos\varphi \cdot \sin\lambda \\ Z &= ((b/a)^2 \cdot \nu + h) \cdot \sin\varphi \end{aligned} \quad (3.4)$$

ECEF coordinates conversion to Geodetic is a tricky process, but EPSG:9602 proposes iteration-less formulas:

$$\begin{aligned} \varphi &= \arctan \frac{Z + \varepsilon \cdot b \cdot \sin^3 q}{p - e^2 \cdot a \cdot \cos^3 q} \\ \lambda &= \arctan \frac{Y}{X} \end{aligned} \quad (3.5)$$

where $\varepsilon = e^2/(1 - e^2)$ – second excentricity, $p = \sqrt{X^2 + Y^2}$ and $q = \arctan \frac{Z \cdot a}{p \cdot b}$. Note, that `atan2(Y, X)` should be used for λ computation.

Ellipsoid	$a = R_{maj}$	$b = R_{min}$	$f = (a - b)/a$	$e = \sqrt{1 - (b/a)^2}$
WGS-84	6378137	6356752.3142	1/298.257223563	0.0818191909289069

Table 3.1: Global ellipsoids paramaters

Rotation from Topocentric (*ENU*) frame to ECEF at specified coordinates $\vec{G}_t$ is done by EPSG:9836:

$$[R_{te}(\varphi, \lambda)] = \begin{bmatrix} -\sin \lambda & -\sin \varphi \cos \lambda & \cos \varphi \cos \lambda \\ \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi \sin \lambda \\ 0 & \cos \varphi & \sin \varphi \end{bmatrix} \quad (3.6)$$

To find geodetic coordinates of the point with *ENU*-coordinates relative to $\vec{G}_0$, it needed to be converted to ECEF $[X_0, Y_0, Z_0]^T$ and:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = [R_{te}(\varphi_0, \lambda_0)] \cdot \begin{bmatrix} E \\ N \\ U \end{bmatrix} + \begin{bmatrix} X_0 \\ Y_0 \\ Z_0 \end{bmatrix} \quad (3.7)$$

Let conversion (3.4) will be $\vec{E} = toECEF(\vec{G})$, conversion (3.5) will be $\vec{G} = fromECEF(\vec{E})$ and (3.6) will be $[R_{te}] = [rotECEF(\vec{G})]$. Then:

$$\vec{G}_t = fromECEF(toECEF(\vec{G}_v) + rotECEF(\vec{G}_v) \cdot [F] \cdot \vec{N}_{vt}) \quad (3.8)$$

where $\vec{G}_t$ – geodetic target coordinates, $\vec{G}_v$ – geodetic Vessel-Frame origin coordinates, $\vec{N}_{vt}$ – target *NED*-coordinates in Vessel-Frame and $[F]$ is flip-matrix for *NED* $\rightarrow$ *ENU* conversion:

$$[F] = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad (3.9)$$

3.3 Final algorithm

The combination of (1.2), (1.3) and (3.8) with assumption, that GNSS antenna is located in Vessel-Frame origin is:

$$\begin{aligned} \vec{X}_{vt} &= \vec{X}_{vu} + [R_{vu}] \cdot [F] \cdot \vec{X}_{ut}, \\ \vec{N}_{vt} &= [R_{an}] \cdot [R_{va}]^T \cdot \vec{X}_{vt}, \\ \vec{G}_t &= fromECEF(toECEF(\vec{G}_v) + rotECEF(\vec{G}_v) \cdot [F] \cdot \vec{N}_{vt}), \end{aligned} \quad (3.10)$$

where:

- $\vec{X}_{ut}$ – coordinates of target extracted from **USBL**LONG;
- $\vec{X}_{vu}$ – USBL offset in Vessel-Frame;
- $[R_{vu}]$ – rotation of USBL-Frame in Vessel-Frame;
- $[R_{an}]$ – rotation of AHRS-frame to NED-frame;
- $[R_{va}]$ – rotation AHRS in Vessel-Frame;
- $\vec{G}_v$ – geodetic coordinates of Vessel-Frame origin.

Note the $[F]$ application to $\vec{X}_{ut}$. In deference from (1.2), axes difference (Fig.2.1) is taken into account.

4 Gravity vector

Gravity vector can be computed in body frame using Euler angles:

$$\vec{g}_a = \vec{g}(\varphi, \theta) = \begin{bmatrix} -\sin \theta \\ \sin \varphi \cos \theta \\ \cos \varphi \cos \theta \end{bmatrix}. \quad (4.1)$$

The property of this vector is that it depends on Roll and Pitch angles only and points downwards:

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T = [R_{an}] \cdot \vec{g}_a. \quad (4.2)$$